

Media Files & Foreign Key Relationships Guide

Step-by-Step Backend Implementation Guide using Django Forms, ORM, and Static Assets Handling

STEP 1: Define Relational Database Schemas (models.py)

```
from django.db import models

class Category(models.Model):
    name = models.CharField(max_length=30)
    details = models.CharField(max_length=300)

    def __str__(self):
        return self.name

class Article(models.Model):
    title = models.CharField(max_length=30)
    desc = models.CharField(max_length=300)
    category = models.ForeignKey(Category, on_delete=models.CASCADE, null=True, blank=True)
    price = models.IntegerField()
    image = models.ImageField(upload_to='sakshi_images', null=True, blank=True)
    video = models.FileField(upload_to='sakshi_videos', null=True, blank=True)
    is_deleted = models.BooleanField(default=False, null=True, blank=True)
    deleted_at = models.DateTimeField(auto_now_add=True, null=True, blank=True)

    def __str__(self):
        return self.title
```

STEP 2: Install Image Library and Execute Database Migrations

```
pip install Pillow
python manage.py makemigrations
python manage.py migrate
```

STEP 3: Configure Media Storage Locations (settings.py)

```
MEDIA_URL = '/media/'
MEDIA_ROOT = BASE_DIR / 'media'
```

STEP 4: Map Project Environment Asset Delivery Routes (urls.py)

```
from django.contrib import admin
from django.urls import path, include
from django.conf import settings
from django.conf.urls.static import static

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('authen.urls')),
    path('base/', include('base.urls'))
```

```
]

if settings.DEBUG:
    urlpatterns += static(settings.MEDIA_URL, document_root=settings.MEDIA_ROOT)
```

STEP 5: Construct Form Validation Architecture (forms.py)

```
from django import forms
from .models import Category, Article

class CategoryForm(forms.ModelForm):
    class Meta:
        model = Category
        fields = '__all__'

class ArticleForm(forms.ModelForm):
    class Meta:
        model = Article
        fields = '__all__'
```

STEP 6: Program View Controllers and Object Operations (views.py)

```
from django.shortcuts import render, redirect, get_object_or_404
from .models import Article, Category
from .forms import ArticleForm, CategoryForm
from django.utils import timezone

def create_category(request):
    form = CategoryForm(request.POST or None)
    if form.is_valid():
        form.save()
        return redirect('category_list')
    return render(request, 'create_category.html', {'form': form})

def category_list(request):
    data = Category.objects.all()
    return render(request, 'category_list.html', {'data': data})

def delete_category(request, pk):
    data = get_object_or_404(Category, id=pk)
    if request.method == 'POST':
        data.delete()
        return redirect('category_list')
    return render(request, 'delete_category.html', {'data': data})

def home(request):
    if request.method == "POST":
        form = ArticleForm(request.POST, request.FILES)
        if form.is_valid():
            form.save()
            return redirect('article_list')
    else:
        form = ArticleForm()
    return render(request, 'home.html', {'form': form})
```

```

def article_list(request):
    data = Article.objects.filter(is_deleted=False)
    return render(request, 'article_list.html', {'data': data})

def update_article(request, pk):
    data = get_object_or_404(Article, id=pk)
    if request.method == "POST":
        form = ArticleForm(request.POST, request.FILES, instance=data)
        if form.is_valid():
            form.save()
            return redirect('article_list')
    else:
        form = ArticleForm(instance=data)
    return render(request, 'update_article.html', {'form': form})

def delete_article(request, pk):
    data = get_object_or_404(Article, id=pk)
    if request.method == 'POST':
        data.is_deleted = True
        data.deleted_at = timezone.now()
        data.save()
        return redirect('article_list')
    return render(request, 'delete_article.html', {'data': data})

def article_history(request):
    data = Article.objects.filter(is_deleted=True)
    return render(request, 'article_history.html', {'data': data})

def article_restore(request, pk):
    data = get_object_or_404(Article, id=pk, is_deleted=True)
    data.is_deleted = False
    data.deleted_at = None
    data.save()
    return redirect('article_history')

```

STEP 7: Establish App Level Application Routes (urls.py)

```

from django.urls import path
from base import views

urlpatterns = [
    path('', views.home, name='home'),
    path('create_category/', views.create_category, name='create_category'),
    path('category_list/', views.category_list, name='category_list'),
    path('delete_category/<int:pk>/', views.delete_category, name='delete_category'),
    path('article_list/', views.article_list, name='article_list'),
    path('update_article/<int:pk>/', views.update_article, name='update_article'),
    path('delete_article/<int:pk>/', views.delete_article, name='delete_article'),
    path('article_history/', views.article_history, name='article_history'),
    path('article_restore/<int:pk>/', views.article_restore, name='article_restore')
]

```

STEP 8: Frontend Submission Structure (create_category.html)

```
{% extends 'main.html' %}
{% block content %}
<h1>Add category</h1>
<a href="{% url 'category_list' %}">View All Category</a>

<form method="POST">
    {% csrf_token %}
    {{ form.as_p }}
    <button type="submit">Submit</button>
</form>
{% endblock %}
```

STEP 9: Category Matrix Visualization Screen (category_list.html)

```
{% extends 'main.html' %}
{% block content %}
<h1>Category List</h1>
<a href="{% url 'create_category' %}">Add Category</a>

<table border="1">
    <tr>
        <th>SL No</th>
        <th>Name</th>
        <th>Details</th>
        <th>Actions</th>
    </tr>
    {% for i in data %}
    <tr>
        <td>{{ forloop.counter }}</td>
        <td>{{ i.name }}</td>
        <td>{{ i.details }}</td>
        <td><a href="{% url 'delete_category' i.id %}">Delete</a></td>
    </tr>
    {% endfor %}
</table>
{% endblock %}
```

STEP 10: Structural Category Deletion Interface (delete_category.html)

```
{% extends 'main.html' %}
{% block content %}
<h1>Are you sure you want to delete the category {{ data.name }}?</h1>
<form method="POST">
    {% csrf_token %}
    <button type="submit">Yes, Delete</button>
    <a href="{% url 'category_list' %}">Cancel</a>
</form>
{% block content %}
```

STEP 11: File Payload Composition Interface (home.html)

```
{% extends 'main.html' %}
{% block content %}
<h1>Add Article</h1>
<a href="{% url 'article_list' %}">View All Articles</a>

<form method="POST" enctype="multipart/form-data">
    {% csrf_token %}
    {{ form.as_p }}
    <button type="submit">Submit</button>
</form>
{% endblock %}
```

STEP 12: Object Collection Visual Interface (article_list.html)

```
{% extends 'main.html' %}
{% block content %}
<h1>Article List</h1>
<a href="{% url 'home' %}">Add Article</a>
<a href="{% url 'article_history' %}">View Deleted Articles</a>

<table border="1">
  <tr>
    <th>SL No</th>
    <th>Title</th>
    <th>Category</th>
    <th>Price</th>
    <th>Description</th>
    <th>Image</th>
    <th>Video</th>
    <th>Actions</th>
  </tr>
  {% for i in data %}
  <tr>
    <td>{{ forloop.counter }}</td>
    <td>{{ i.title }}</td>
    <td>{{ i.category }}</td>
    <td>{{ i.price }}</td>
    <td>{{ i.desc }}</td>
    <td>
      {% if i.image %}
        
      {% endif %}
    </td>
    <td>
      {% if i.video %}
        <video src="{{ i.video.url }}" width="150px" controls>
          Your browser does not support the video tag.
        </video>
      {% else %}
        No Video
      {% endif %}
    </td>
    <td>
      <a href="{% url 'update_article' i.id %}">Edit</a>
      <a href="{% url 'delete_article' i.id %}">Delete</a>
    </td>
  </tr>
  {% endfor %}
</table>
{% endblock %}
```

```

        </td>
    </tr>
    {% endfor %}
</table>
{% endblock %}

```

STEP 13: Instance Updating Form Interface (update_article.html)

```

{% extends 'main.html' %}
{% block content %}
<h1>Update Article</h1>
<a href="{% url 'article_list' %}">Back</a>

<form method="POST" enctype="multipart/form-data">
    {% csrf_token %}
    {{ form.as_p }}
    <button type="submit">Update</button>
</form>
{% endblock %}

```

STEP 14: Soft Deletion Object Confirmation (delete_article.html)

```

{% extends 'main.html' %}
{% block content %}
<h1>Are you sure you want to delete the article {{ data.title }}?</h1>
<form method="POST">
    {% csrf_token %}
    <button type="submit">Yes, Delete</button>
    <a href="{% url 'article_list' %}">Cancel</a>
</form>
{% endblock %}

```

STEP 15: Soft Deleted Records Dashboard (article_history.html)

```

{% extends 'main.html' %}
{% block content %}
<h1>Article History</h1>
<a href="{% url 'article_list' %}">Back</a>

<table border="1">
    <tr>
        <th>SL No</th>
        <th>Title</th>
        <th>Category</th>
        <th>Price</th>
        <th>Description</th>
        <th>Image</th>
        <th>Video</th>
        <th>Deleted Time</th>
        <th>Actions</th>
    </tr>
    {% for i in data %}
    <tr>

```

```

        <td>{{ forloop.counter }}</td>
        <td>{{ i.title }}</td>
        <td>{{ i.category }}</td>
        <td>{{ i.price }}</td>
        <td>{{ i.desc }}</td>
        <td>
            {% if i.image %}
                
            {% endif %}
        </td>
        <td>
            {% if i.video %}
                <video src="{{ i.video.url }}" width="150px" controls>
                    Your browser does not support the video tag.
                </video>
            {% else %}
                No Video
            {% endif %}
        </td>
        <td>{{ i.deleted_at }}</td>
        <td><a href="{% url 'article_restore' i.id %}">Restore</a></td>
    </tr>
    {% endfor %}
</table>
{% endblock %}

```

STEP 16: Navigation Elements Layout (nav.html)

```

{% if request.user.is_authenticated %}
    <a href="{% url 'category_list' %}">Category List</a>
    <a href="{% url 'article_list' %}">Article List</a>
    <a href="#">Profile</a>
    <a href="#">Logout</a>
{% else %}
    <a href="#">Register</a>
    <a href="#">Login</a>
{% endif %}

```